Section 13.2: Lenz's Law

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A. Answers may vary. Sample answer:
The direction of the current in Step 3 was opposite to the direction of the current in Step 2.
B. Answers may vary. Sample answer:
When repeating Step 2 with the south pole of the magnet, the direction of the current was opposite to the direction when the north pole of the magnet was used. When repeating Step 3 with the south pole of the magnet, the direction of the current was again opposite to the direction when the north pole of the magnet was used.
C. Using the south pole instead of the north pole changed the order in which the direction of the current changed. It did not affect how quickly the current changed or how much current was produced.
D. Using the right-hand rule for a coiled conductor, I predicted that the end of the coil nearest to the magnet was a north magnetic pole in Step 2 and a south magnetic pole in Step 3.

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1. (a) The north pole of a permanent magnet is being forced into the coil. By Lenz's Law, the coil must oppose that with a north pole on the right side of the coil. If the right side of the coil is north, then the left side of the coil must be south. The right-hand rule for a solenoid determines the direction of the electric current to be down the coil at the front.
(b) The north pole of a permanent magnet is being pulled out of the coil. By Lenz's Law, the coil must oppose the north pole with a south pole on the right side of the coil. If the right side of the coil is south, then the left side of the coil must be north The right-hand rule for a solenoid determines the direction of the electric current to be up the coil at the front.
(c) The south pole of a permanent magnet is being pulled out of the coil. By Lenz's Law, the coil must oppose the south pole with a north pole on the right side of the coil. If the right side of the coil is north, then the left side of the coil must be south. The right-hand rule for a solenoid determines the direction of the electric current to be down the coil at the front.
(d) The south pole of a permanent magnet is being forced into the coil. By Lenz's Law, the coil must oppose the south pole with a south pole on the right side of the coil. If the right side of the coil must be south, then the left side of the coil must be north. The right-hand rule for a solenoid determines the direction of the electric current to be up the coil at the front.
2. The answers to Question 1 would not change if the coils were moved instead of the magnets, because the coils would experience a changing magnetic field that is the same as if the magnets are moved and the coils are stationary. By Lenz's law, the electric current in each coil would have to be in a direction such that its own magnetic field opposes the change that produced it, which in this case is the movement of the coil itself. For this to happen, each coil must have the same direction of induced current and magnetic poles as in Question 1.
3. When a magnet is pushed into a coil, the kinetic energy of the magnet is transformed into electrical energy in the electric current in the coil. If the magnetic field that is created by the induced current were to attract the magnet into the coil, both the kinetic energy of the magnet and the electrical energy in the electric current would increase without any work being done by the person pushing the magnet. This increase in energy of the system without any work being done would violate the law of conservation of energy.
4. Using electromagnets in the carts instead of permanent magnets would work equally well since the braking system would still function in the same way. It would not be as reliable because electromagnets can only generate a magnetic field when they are connected to a power source, so if the power failed the electromagnets would not produce a magnetic field and the braking system would not work. Permanent magnets always produce a magnetic field so they avoid this problem.